



ICHTHYOSAUR FOSSIL LOCALITY INTELLIGENCE REPORT

Humboldt Range & Gabbs Valley — Central Nevada

Generated March 2026	Model Version RF-v1	Cross-Val AUC 0.906	Candidate Sites 50 Targets	Priority #1 Score 1.000
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MISSION: Identify high-probability ichthyosaur fossil localities in central Nevada using machine learning on USGS terrain data, cross-validated against PBDB occurrence records and Nevada Triassic marine geology. This report provides actionable field targets ranked by composite probability score.

■ IMPORTANT DISCLAIMERS & SAFETY INFORMATION

RESEARCH TOOL — NOT A GUARANTEE

This report is generated by a machine learning model trained on publicly available terrain and paleontological data. Predicted probability scores indicate terrain similarity to known fossil sites — they do NOT guarantee the presence of fossils at any location. Many high-scoring sites will yield no fossils. Field results will vary significantly.

KNOW THE LAW BEFORE YOU GO

Fossil collection laws in the United States are complex and vary by land ownership. On BLM and U.S. Forest Service land, casual collection of reasonable quantities of common invertebrate and plant fossils for personal use is generally permitted without a permit. HOWEVER: Collection of vertebrate fossils (including ichthyosaurs and all other reptile, fish, and mammal remains) on federal land requires a Paleontological Resources Preservation Act (PRPA) permit. Unpermitted collection of vertebrate fossils on federal land is a federal crime punishable by fines up to \$20,000 and/or imprisonment. Contact the BLM Winnemucca District Office (775-623-1500) BEFORE collecting any vertebrate material.

VERIFY LAND OWNERSHIP

Target coordinates are based on geographic analysis only. Land ownership has NOT been verified for individual sites. Sites may fall on BLM, U.S. Forest Service, tribal, state, or private land with different regulations. Always verify using BLM GeoCommunicator before entering any property. Trespassing on private or tribal land is illegal.

REMOTE TERRAIN & PERSONAL SAFETY

Target sites are in extremely remote, rugged terrain with no cell service, no water sources, extreme temperature swings, and flash flood risk. Do not go alone. Always carry a satellite communicator (e.g. Garmin inReach), minimum 4L water per person per day, first aid kit, and paper topo map. Inform someone of your route before departure. The authors accept no liability for injury, death, or legal consequences arising from use of this document.

SITE DISTURBANCE & SCIENTIFIC INTEGRITY

Even where collection is legal, unnecessary disturbance degrades scientific value. Photograph everything in place before moving it. Record precise GPS coordinates for any finds. Report significant discoveries to the Nevada State Museum and consider contributing to the Paleobiology Database (paleobiodb.org).

MODEL LIMITATIONS

This model uses terrain data only. It does not account for subsurface geology, fault displacement, erosion history, or prior collection effort. Training dataset contains only 27 positive examples. Treat predictions as one input among many — consult a professional geologist or paleontologist before making significant decisions based on this output.

By using this report you acknowledge these disclaimers and accept full personal responsibility for compliance with all applicable laws, regulations, and safety precautions.

MODEL SUMMARY

Metric	Value	Notes
Training positives	27	PBDB ichthyosaur occurrences, Nevada
Background points	270	Random terrain samples, 10:1 ratio
Algorithm	Random Forest	500 trees, max depth 6, balanced classes
Cross-validation AUC	0.906 +/- 0.072	5-fold stratified CV
Top feature	Slope (0.293)	Closely followed by TRI (0.291)
DEM source	USGS 3DEP 1/3 arc-second	~10m resolution, 6 tiles merged
Study area	700M+ pixels	n39-41, w117-119 coverage
Candidate sites	50	Top-ranked from full prediction surface
Geology source	USGS NGMDB Nevada 1:500k	Triassic marine units filtered

FEATURE IMPORTANCE

Terrain metrics ranked by Random Forest importance. Slope and ruggedness together explain 58% of predictive power — fossil sites favor moderately inclined, eroding ridge flanks.

Feature	Importance	Interpretation
Slope (degrees)	0.293	Moderate 8-15 degree slopes favor exposed outcrops
Ruggedness (TRI)	0.291	Low-moderate roughness — eroding but accessible
Aspect (degrees)	0.222	Both N and S facing slopes represented
Elevation (m)	0.195	Sweet spot 1400-1750m, mean 1647m

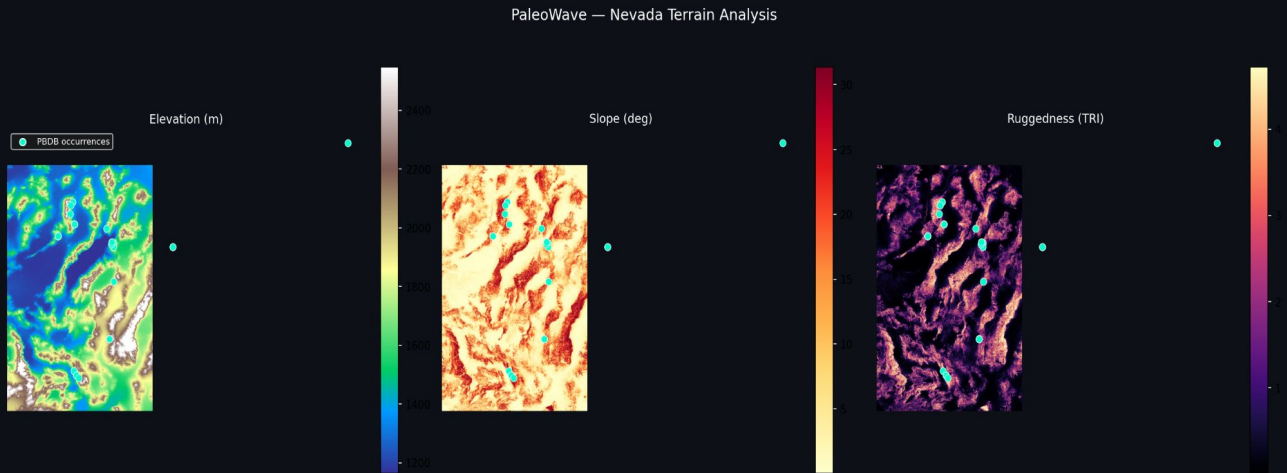


Figure 1: Nevada terrain analysis — elevation, slope, and ruggedness with PBDB occurrences (teal dots).

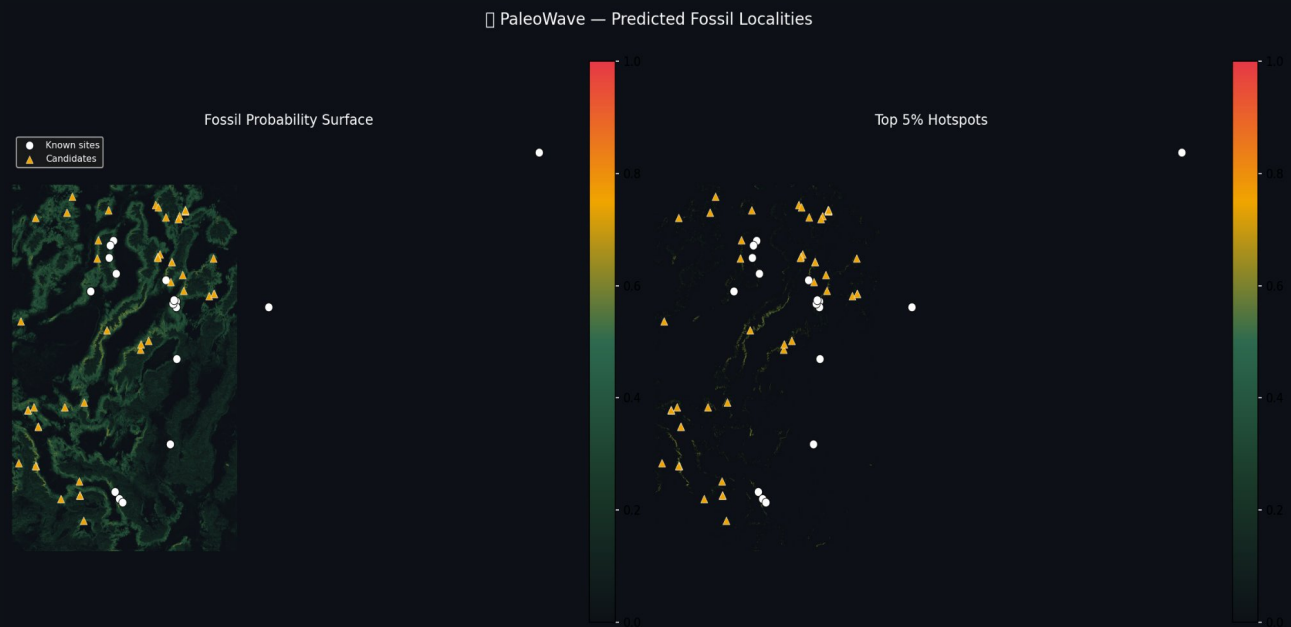
TOP 10 PRIORITY FIELD TARGETS

Ranked by composite score: ML terrain probability + geology bonus for proximity to TRc (Triassic limestone/dolomite — Favret, Luning, Star Peak Group, Augusta Mountain). Vertebrate fossil collection requires PRPA permit on federal land — see disclaimer.

#1	40.404722N, -118.243889W 40° 24' 17.0" N 118° 14' 38.0" W	ML Probability: 0.8005 Geology: ON TRc formation (52m) Formation: TRc — Triassic limestone/dolomite	1.000 0
#2	40.209167N, -117.587778W 40° 12' 33.0" N 117° 35' 16.0" W	ML Probability: 0.7952 Geology: 992m to TRc formation Formation: TRc — Triassic limestone/dolomite	0.895 2
#3	40.550278N, -118.232500W 40° 33' 1.0" N 118° 13' 57.0" W	ML Probability: 0.7946 Geology: 1361m to TRc formation Formation: TRc — Triassic limestone/dolomite	0.894 6
#4	40.434444N, -117.686667W 40° 26' 4.0" N 117° 41' 12.0" W	ML Probability: 0.8423 Geology: 4781m to TRc formation Formation: TRc — Triassic limestone/dolomite	0.892 3
#5	40.839722N, -117.720833W 40° 50' 23.0" N 117° 43' 15.0" W	ML Probability: 0.7920 Geology: 1059m to TRc formation Formation: TRc — Triassic limestone/dolomite	0.892 0
#6	40.822778N, -117.696944W 40° 49' 22.0" N 117° 41' 49.0" W	ML Probability: 0.7899 Geology: 1309m to TRc formation Formation: TRc — Triassic limestone/dolomite	0.889 9
#7	40.418889N, -117.704444W 40° 25' 8.0" N 117° 42' 16.0" W	ML Probability: 0.8305 Geology: 3338m to TRc formation Formation: TRc — Triassic limestone/dolomite	0.880 5
#8	40.787222N, -117.456389W 40° 47' 14.0" N 117° 27' 23.0" W	ML Probability: 0.8533 Geology: 13040m to TRc formation Formation: TRc — Triassic limestone/dolomite	0.853 3
#9	40.907222N, -118.464444W 40° 54' 26.0" N 118° 27' 52.0" W	ML Probability: 0.8523 Geology: 41459m to TRc formation Formation: TRc — Triassic limestone/dolomite	0.852 3
#10	40.113056N, -117.202500W 40° 6' 47.0" N 117° 12' 9.0" W	ML Probability: 0.7991 Geology: 4801m to TRc formation Formation: TRc — Triassic limestone/dolomite	0.849 1

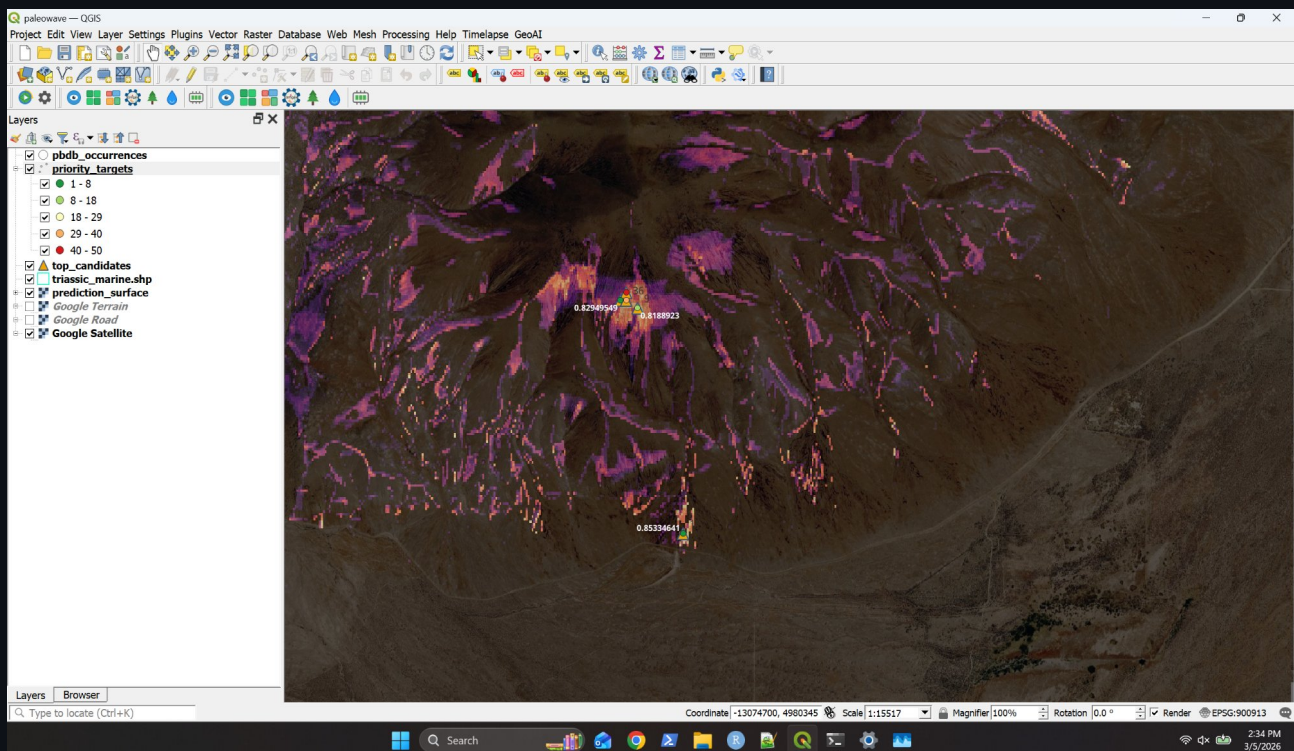
PREDICTION SURFACE & HOTSPOT MAP

Left: Full fossil probability surface. Right: Top 5% hotspots. White dots = known PBDB occurrences. Gold triangles = ML candidate sites.



PRIMARY TARGET ZONE — HUMBOLDT RANGE

Priority #1 (40.4047N, 118.2439W) sits 52m from confirmed TRc Triassic marine limestone. Purple/white = ML probability. Cyan outlines = Triassic formations. Dots = priority targets 1-50.



FIELD PREPARATION NOTES

Access

Humboldt Range targets accessed via US-95 north from Lovelock, then east on dirt roads. High-clearance 4WD required. Check BLM road conditions. Verify land status before entering.

Best Season

April-June and September-October. Avoid July-August (38C+) and January-February snow. Spring provides best erosional surface exposure after freeze-thaw cycles.

What to Look For

Grey to buff Triassic limestone and dolomite outcrops, often platy or nodular. Dark grey chert nodules characteristic of Favret and Prida formations. Bone material appears as darker, denser inclusions with subtle texture differences.

Permits — READ THIS

Vertebrate fossil collection on federal land requires a PRPA permit. Contact BLM Winnemucca District (775-623-1500) BEFORE collecting vertebrate material. Unpermitted collection is a federal crime.

Safety Equipment

GPS with loaded coordinates, rock hammer, hand lens (10x min), sample bags, field notebook, camera, topo maps, satellite communicator, 4L+ water/person/day, first aid kit.

Reporting Finds

Report significant finds to Nevada State Museum. Contribute locality data to Paleobiology Database (paleobiodb.org) to improve future research.

QUICK REFERENCE — TARGET COORDINATES

Load priority_targets.geojson into GPS or mapping app for all 50 sites.

#	Decimal Degrees	DMS Latitude	DMS Longitude	Score
1	40.404722, -118.243889	40° 24' 17.0" N	118° 14' 38.0" W	1.0000
2	40.209167, -117.587778	40° 12' 33.0" N	117° 35' 16.0" W	0.8952
3	40.550278, -118.232500	40° 33' 1.0" N	118° 13' 57.0" W	0.8946
4	40.434444, -117.686667	40° 26' 4.0" N	117° 41' 12.0" W	0.8923
5	40.839722, -117.720833	40° 50' 23.0" N	117° 43' 15.0" W	0.8920
6	40.822778, -117.696944	40° 49' 22.0" N	117° 41' 49.0" W	0.8899
7	40.418889, -117.704444	40° 25' 8.0" N	117° 42' 16.0" W	0.8805
8	40.787222, -117.456389	40° 47' 14.0" N	117° 27' 23.0" W	0.8533
9	40.907222, -118.464444	40° 54' 26.0" N	118° 27' 52.0" W	0.8523
10	40.113056, -117.202500	40° 6' 47.0" N	117° 12' 9.0" W	0.8491